

Standard Operating Procedure

KT-400 Controller — Default & Configuration Procedure

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Related Documents	SOP-LV-004, SOP-LV-013, SOP-LV-014
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Applies To	Field Technicians, Installers
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Owner	Apex – Operations
Approved By	_____

1. Purpose

This procedure defines the standard method for resetting a Kantech KT-400 door controller to factory default and configuring it for use on an Apex-installed access control system, ensuring consistent commissioning and reducing field errors.

2. Scope

Applies to all Apex technicians installing, replacing, or troubleshooting Kantech KT-400 controllers on EntraPass Special, Corporate, or Global Edition systems.

3. Regulatory & Reference Standards

- Ontario Occupational Health and Safety Act (OHSA) and O. Reg. 213/91 – Construction Projects
- Manitoba Workplace Safety and Health Act
- CSA C22.1 – Canadian Electrical Code, low-voltage wiring practices
- Manufacturer reference: Kantech KT-400 Ethernet Four-Door Controller Installation Manual, DN2003-1405 (verified)

4. Tools & Materials Required

- Laptop with KT-Finder installed and EntraPass workstation access
- Ethernet patch cable
- Small flathead screwdriver (for jumper access)
- Site IP addressing scheme / network configuration sheet
- PPE per SOP-LV-004

5. Factory Default Reset

The KT-400 uses jumpers JP2/JP3 in combination with the Reset button.

1. Before resetting, confirm: Ethernet is connected to the network (green LINK LED lit, yellow TXRX LED flashing) and the blue Heartbeat LED is flashing.
2. Remove jumpers JP2 and JP3.
3. Press the Reset button.
4. Check the blue Heartbeat LED pattern to confirm factory default DHCP mode.
5. Verify the controller's IP address using KT-Finder.
6. Reinstall jumpers JP2 and JP3.
7. Proceed to configuration using KT-Finder.

KT-400 Connection Reference

KT-400 — Connection Reference (Functional Groups)

Terminal names and functions per Kantech KT-400 Installation Manual (DN2003-1405)
This is a functional reference, not an exact physical PCB layout — confirm exact terminal position against the board silkscreen.



RS-485 loop wiring, termination, and topology: see SOP-LV-013. Fire alarm / life-safety lock release requirements: see SOP-LV-014.

- Lock outputs (LK1– through LK4–, shared +) are open-drain and supervised — one per onboard door.
- Zone inputs Z1–Z16 default to: Z1/Z2 = Door 1 contact/REX, Z5/Z6 = Door 2, Z9/Z10 = Door 3, Z13/Z14 = Door 4, with the remaining zones available as general-purpose inputs.
- RS-485 signal ground uses the 12VDC AUX – (or another GND terminal) — never the EGND (earth ground) terminal, which is for chassis/earth bonding only.
- No end-of-line resistor is installed on the RS-485 (COM1) terminals of the last KT-400 in a loop.
- RS-485 loop wiring, termination, and topology: see SOP-LV-013. Fire alarm / life-safety lock release requirements: see SOP-LV-014.

6. Configuration (EntraPass)

- Connect the laptop to the same network/subnet as the controller and open KT-Finder.
- Confirm the controller appears in the list, identified by MAC address.
- Assign a static IP address per the site's IP addressing scheme (recommended for EntraPass-managed controllers).
- Set the operating mode to EntraPass mode (not Standalone/Cloud) unless the site specifically requires standalone operation.
- In EntraPass, go to Devices → Connection configuration and add/confirm the connection.
- In EntraPass, go to Devices → Configuring controllers, add the controller, select model KT-400, and enter its IP/MAC address.
- Reload the controller (Operations → Manual operations on controllers → Reload) to push door, input, and relay configuration to the unit.

7. Verification

- Controller shows Online status in EntraPass
- Test at least one door, one input, and one relay before terminating remaining field wiring
- Confirm event messages appear correctly in the EntraPass message desktop
- Record controller MAC address, assigned IP, and location on the site network configuration sheet

8. Notes

If the controller fails to appear in KT-Finder after a factory default reset, confirm the laptop and controller are on the same broadcast domain/subnet, and that no managed switch port security is blocking DHCP broadcast traffic.

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